

Activity 1 — Rewriting a Vague Problem into a Measurable Problem Statement

ShopFront SG — 'Customers are unhappy and sales are dropping'

Topic	Topic 1
Objective	A1 · K1 — Identify the type of performance deficiency and express it as a measurable problem statement.
Duration	45 minutes
Grouping	Teams of 3
Tools	ChatGPT / Copilot / Gemini, CollabNote
Ed-tool	CollabNote — https://alfredang.github.io/collabnote/

The Scenario

ShopFront SG is a 42-outlet fashion retailer with an e-commerce site and a mobile app. At the Monday management meeting the Head of Retail Operations opens with one line: "Customers are unhappy and sales are dropping — fix it." No metric, no baseline, no scope. Three departments leave the room with three different interpretations. Marketing books a S\$180,000 discount campaign. IT starts an app redesign. Customer Service hires two temps. Ninety days later the complaint rate is unchanged and the budget is gone.

You are the business analyst asked to restate the problem BEFORE any more money is committed. The dashboard extract below is all the evidence you have.

The Evidence

ShopFront SG — 90-day performance extract

Metric	12 months ago	Current	Industry benchmark
Average app page load	2.1 s	4.8 s	under 2.0 s
Checkout completion rate	68%	51%	70%
Cart abandonment	61%	74%	60%
CSAT (post-purchase)	84%	70%	85%
Repeat purchase (90-day)	38%	29%	40%
Complaint tickets / week	120	410	-
In-store sales	S\$4.2m/mo	S\$4.1m/mo	-
Online sales	S\$2.8m/mo	S\$1.9m/mo	-

Discussion Questions

1. Which metric in the table is the SYMPTOM the executive noticed, and which metric is closest to the actual performance deficiency? Justify with the numbers.
2. The evidence shows in-store sales are flat while online sales fell 32%. What does that single contrast rule out as a cause, and why does that matter before you spend a dollar?
3. Write the problem statement with all six elements. Which element was hardest to fill, and what does the gap tell you about your organisation's data?
4. Compare your team's statement with the GenAI-generated one. What did the AI assume that is NOT supported by the evidence? (Every team will find at least one.)
5. Whose sign-off do you need on this problem statement before work starts, and what would you show them to get it?

Step-by-Step

6. Read the ShopFront SG scenario and the 90-day evidence table as a team.
7. Individually and in silence, underline the one metric you believe is the real performance deficiency. Do not discuss yet — this prevents anchoring on the loudest voice.
8. Compare choices around the table. Where you disagree, argue from the numbers only.
9. Draft the problem statement on paper with all six elements labelled: Context, Metric, Baseline, Target, Timeframe, Constraints.
10. Open your GenAI tool and run the Problem Statement prompt supplied on the slide.
11. Compare the AI output against your draft. Highlight every specific the AI supplied that the evidence did not support — label each one 'ASSUMPTION'.
12. Merge the two into a final statement your team can defend to the Head of Retail Operations.
13. Post the final statement to the shared wall at <https://alfredang.github.io/collabnote/>
14. Nominate one speaker to read it out in 60 seconds and name the one assumption you are least comfortable with.

The GenAI Prompt

You are a business analyst for a Singapore fashion retailer.

Refine the vague statement below into ONE clear, measurable problem statement.

Vague statement: "Customers are unhappy and sales are dropping."

Evidence:

- App page load rose from 2.1s to 4.8s (benchmark <2.0s)
- Checkout completion fell 68% -> 51% (benchmark 70%)
- Cart abandonment rose 61% -> 74%
- CSAT fell 84% -> 70%
- Online sales fell S\$2.8m -> S\$1.9m per month; in-store flat

Your problem statement MUST contain, each labelled:

- Context (who, where, which process)

- Metric (the single primary measure)
- Baseline (where it is now, with the number)
- Target (where it must be, with the number)
- Timeframe (by when)
- Constraints (budget, headcount, compliance, brand)

Then list any assumption you had to make, and state what additional evidence would let you remove that assumption. Do not propose solutions yet.

Trainer Debrief

Expected answers and the points to draw out. Trainer reference.

EXPECTED DIRECTION — the deficiency is a DIGITAL CHANNEL conversion failure, not a general 'unhappy customer' problem. In-store being flat while online fell 32% localises the problem to the digital funnel and rules out brand, product and pricing as the primary cause — which immediately invalidates Marketing's S\$180k discount campaign.

A defensible statement reads roughly: 'Checkout completion on the ShopFront SG app and web store has fallen from 68% to 51% (benchmark 70%) over 12 months, alongside app load times rising from 2.1s to 4.8s, contributing to a S\$0.9m per month decline in online sales. We aim to restore checkout completion to at least 68% and app load to under 2.0s within 90 days, without additional headcount and without changes requiring PDPA re-consent.'

KEY TEACHING POINTS:

1. The executive named a symptom ('unhappy'). CSAT is a lagging perception measure; checkout completion is the operational deficiency you can actually act on.
2. Six elements are non-negotiable. A statement missing a Baseline cannot be evaluated later — this is exactly what Topic 3 comes back to.
3. GenAI reliably invents plausible specifics (a peak-hour window, a payment gateway name, a customer segment) that the evidence never supplied. Learners must mark these as assumptions. This is the single most important AI-literacy habit in the course.
4. Note what has NOT been established yet: WHY load time rose. That is root cause, and it is Activity 2. Resist the team that jumps to 'we need more servers'.

Self-Check — Is Your Output Finished?

Your statement is complete when a colleague who was NOT in the room can read it and correctly state (a) which metric is being fixed, (b) the number it is at today, (c) the number it must reach, (d) by when, and (e) what you are not allowed to change — without asking you a question. If any of the five is missing, the statement is not finished.